Feature Driven K-Means Segmentation of Retinal Cysts in OCT Imaging

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1. Introduction

Retinal cysts have pathological significance in several eye disorders. Detecting and quantifying such cysts from optical coherence tomography (OCT) scans is currently tedious and requires expertise. The precise segmentation and quantification of intraretinal cysts (IRCs) is of high clinical value, primarily because IRCs are key anatomical biomarkers for vision-threatening diseases such as diabetic macular edema, retinal vein occlusion, and age-related macular degeneration [1].

Developing automated algorithms for reliable IRC segmentation is nontrivial and presents several challenges that must be addressed in the pipeline design. OCT images inherently suffer from speckle noise due to the physics of light scattering, which significantly degrades image quality [2]. This noise combined with the fact that cysts are low-reflectance (dark) structures often nestled within tissue of similar intensity, results in low contrast and blurred boundaries. This challenges accurate edge detection, particularly for small cysts.

Algorithms struggle with generalizability due to the wide variation in image appearances stemming from different OCT vendors and acquisition protocols [3]. Additionally, artifacts such as vessel shadowing or imperfect retinal layer segmentation which is required to define the region of interest can lead to significant false positive errors.

The exact morphology of cysts varies widely in size and shape, from small microcystic structures to large cavities with several separate pockets. This high variability complicates fixed-parameter and feature-based segmentation methods, which often fail to detect smaller cysts or generate false positives by labelling some areas as cysts when it shouldn’t.

The segmentation pipeline approach in this report is divided into three main stages; pre-processing and detecting our region of interest (ROI) which in our case is the retina as proposed in Ganjee et al. [4], since all cysts are in the retina. Then the main segmentation using k-means clustering inspired by Girish et al. [1], followed by post processing operations of morphological closing and further filtering to remove false positives.

2. Methodology

As mentioned, the methodology is divided into three sequential steps: pre-processing to produce a denoised image and a region of interest definition, feature-driven K-Means clustering, and post processing for refinement.

The pipeline first isolates the relevant retinal tissue to reduce false positives from surrounding, dark non-tissue areas. First, we apply contrast enhancement using Contrast Limited Adaptive Histogram Equalization (CLAHE), followed by bilateral filtering to reduce noise while preserving edges. Then, Otsu's thresholding (which is an image segmentation method that automatically determines the optimal threshold value by minimizing the variance within each class of pixels) is used to identify the bright retina region followed by morphological closing (using a 20×20 elliptical kernel) to fill the gaps inside the retina mask. At last, connected component analysis is used to keep only the largest connected component which is the retina generating the region of interest mask.

The core segmentation uses K-Mean clustering with (K = 3) to classify pixels based on feature to separate the cysts. We begin by extracting two features for each pixel; intensity, which is every pixel's intensity from the denoised image, and "blobness" which is a morphological feature calculated using the Black Hat operation from the OpenCV library with a 15×15 elliptical kernel. This feature strongly highlights small, dark, circular cysts against a brighter background. After that, the k-means clustering is applied to the featured data initialized using K-Means++ for robust center selection. To identify the cluster that represents the cysts, we calculate a score for each cluster using the formula:

$$score = mean\ intensity - mean\ blobness$$

we want high blobness and low intensity, since the cysts are dark round shapes. Which leads to picking the cluster with the lowest calculated score which forms the segmented cyst mask. A bitwise AND operation is performed with the previously generated Retinal ROI Mask to restrict the segmented area solely to the retinal tissue.

The resulting mask undergoes a series of post processing cleanup steps to ensure morphological accuracy and to remove noise that was falsely identified as cysts. The mask is first smoothed via Morphological Closing (using a 3x3 ellipse kernel) and then undergoes binary hole filling (imported from the SciPy library) to ensure segmented cyst regions are solid structures. Connected Component Analysis is performed again to measure the area of each segmented object. Any object smaller than a certain minimum size threshold is removed, effectively eliminating small noise and reducing false positives. Finally, we perform near-edge removal**,** removing objects that are either located within a 50-pixel margin of the top, left, or right image borders or located close to the bottom border within a 175-pixel margin, as these often correspond to shadowing or vessel artifacts below the retina.

Figure 1 in the next page shows the workflow of the full pipeline.

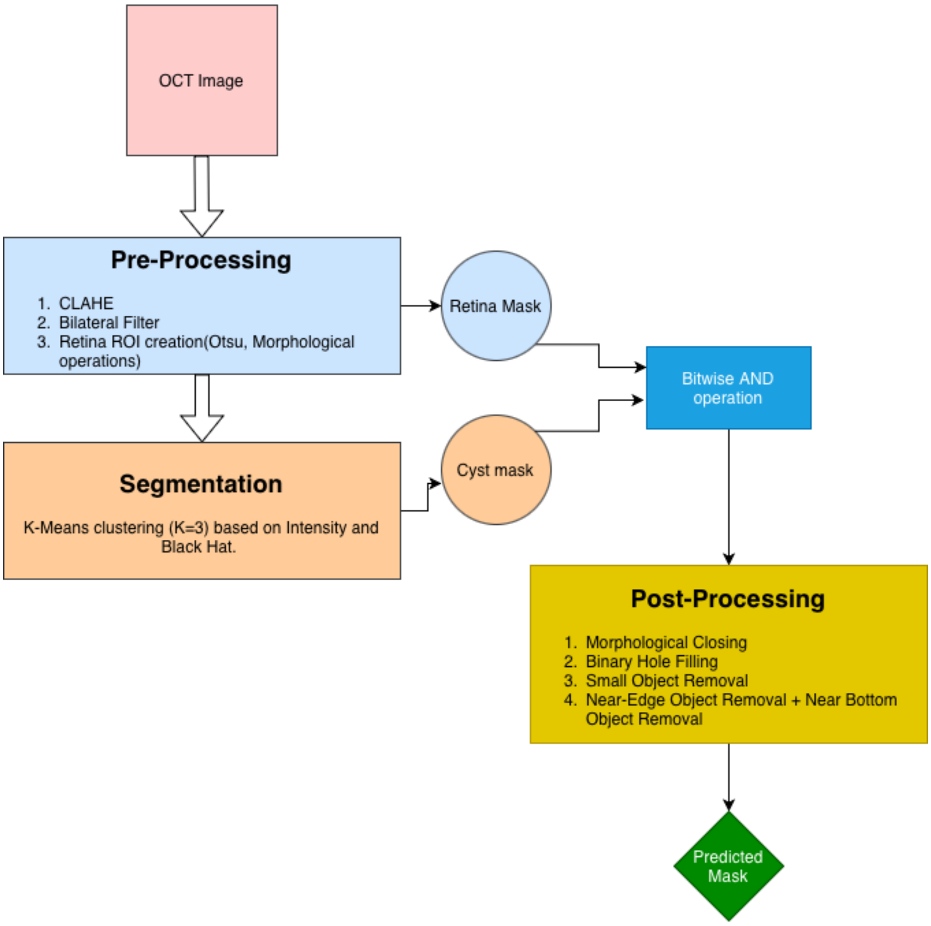

Figure 1: Workflow of the full segmentation pipeline

3. Implementation Details

The pipeline uses several python libraries to handle image processing, analysis, and evaluation. **NumPy** was used for array operations and numerical processing, necessary for manipulating image pixel data as multi-dimensional arrays. **OpenCV (cv2)** supplied core image processing functions, including CLAHE, bilateral filtering, thresholding, morphological operations, and K-Means clustering, with (cv2.kmeans) serving as a critical step in the core segmentation pipeline due to its speed and reliability. **Matplotlib** enabled the visualization of both qualitative and quantitative results, displaying plots and graphs. **Pandas** was used for handling and presenting quantitative evaluation metrics, allowing aggregation and calculation of mean values for IoU, Dice, and other metrics in structured data frames. Advanced spatial image processing tasks were addressed using **SciPy** (ndimage, binary_fill_holes), which provided functionality not available in OpenCV, including connected component analysis and post-processing refinement. Finally, **scikit-image** (skimage.measure) was used for region property analysis, with the (regionprops) function enabling calculation of area, bounding boxes, and other properties, which were vital for ROI selection and post-processing cleanup of artifacts.

The final selection of all pipeline parameters (e.g. K, Epsilon, MIN_SIZE, etc.) was achieved through iterative parameter tuning loops to optimize results on the training dataset.

4. Results

-Qualitative Results:

The qualitative results are represented by a figure that shows the original OCT image, the ground truth annotation, predicted mask and overlay (Green = True Positive, Red = False Positive, Blue = False Negative).

Best result was achieved on image number (2) with an IoU of ~57% and very few false negatives

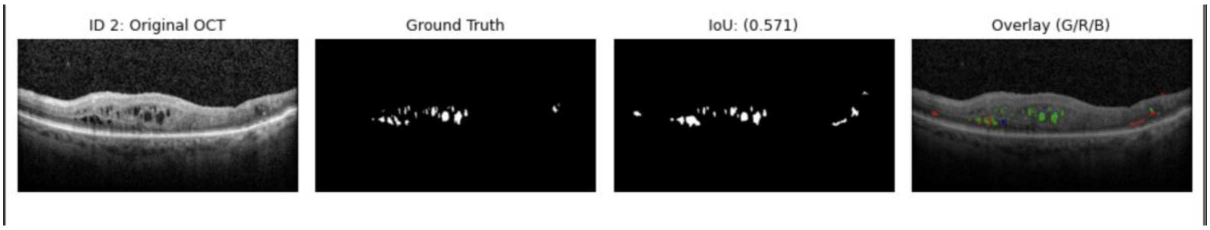

Figure 2: Qualitative results for image (2); best result

The lowest result was for image number (9), where the algorithm produced too many false positives as seen in *Figure 3*

Figure 3: Qualitative results for image (9); worst case

Overall, the mean IoU scored for all ten images in our dataset was around 35%

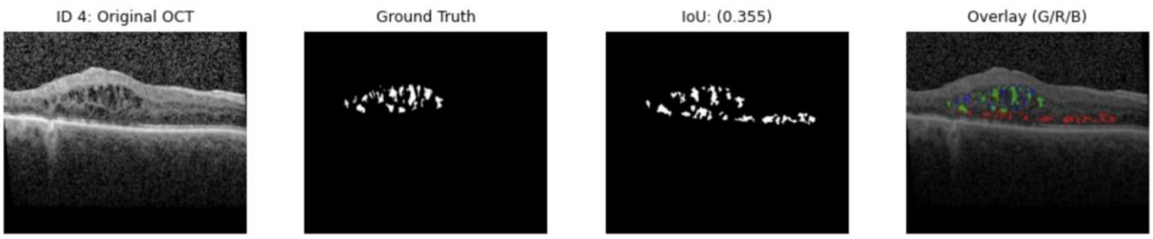

Figure 4: Qualitative results for image (4)

Due to K-Means' sensitivity to the initial centroid placement, each run may converge to a different local optimum, thus causing the reported metrics to vary slightly for each run of code.

-Quantitative Results:

The table below displays the calculated evaluation metrics; Intrsection over Union (IoU), Dice Coefficient, Sensitivity, and Specificity for all ten images and the mean values for each metric.

Image ID	IoU	Dice	Sensitivity	Specificity
1	0.2088	0.3455	0.4093	0.9932
2	0.5712	0.7271	0.8499	0.9958
3	0.2701	0.4253	0.3636	0.9944
4	0.3554	0.5244	0.6445	0.9900
5	0.3088	0.4719	0.4431	0.9934
6	0.5175	0.6820	0.7250	0.9955
7	0.3232	0.4885	0.7250	0.9955
8	0.4264	0.5978	0.5648	0.9937
9	0.2011	0.3349	0.6282	0.9845
10	0.3879	0.5590	0.6003	0.9920
Mean	0.3570	0.5157	0.5932	0.9912

Table 1: Quantitative results for all ten images and mean

The clustered column chart shows the IoU results and the mean IoU

Figure 5: IoU scores column chart

5. Discussion

The segmentation algorithm achieved high specificity of about 99% while scoring way lower on other metrics, meaning that when the algorithm says a pixel is not a cyst, it is highly likely to be correct, but not the other way around as evidenced by the result sensitivity which had a mean of around 59% which is approximately the same percentage in Girish et. All [1] (achieved ~57% Recall which is the same as sensitivity using K-Means)

As for other metrics, the algorithm scored around 35% mean IoU and 51% mean Dice coefficient, the largest source of error identified during the qualitative review is the presence of prominent false positives (red regions in the overlay), which primarily occur due to vessel shadowing and other dark, non-cyst structures (noise) being misclassified. These false positives are critically responsible for lowering the IoU.

The column chart shows that the algorithm's results, as measured by IoU scores, varied substantially from one image to another, highlighting the impact of varying cyst characteristics and image conditions on segmentation performance.

6. Conclusion

This report presented a Feature-Driven K-Means Clustering pipeline for the automated segmentation of intraretinal cysts from OCT scans. The method utilizes a two-dimensional feature space of pixel intensity and Black Hat morphological blobness leveraging comprehensive pre-processing and post-processing.

The pipeline demonstrated high performance in correctly identifying non-cyst regions (true negatives), as evidenced by a high specificity score. This confirms the success of the pre-processing and robust post-processing steps in suppressing background noise and irrelevant structures.

The use of K-Means clustering driven by explicit, domain-specific features provides a simple and computationally efficient approach compared to other complex models. However, the inherent reliance on the K-Means algorithm means the results are non-deterministic and sensitive to initial centroid placement, causing variations between independent runs.

The core limitation of the pipeline is its overall low spatial accuracy driven by a high rate of false positives. The simple two-dimensional feature space is insufficiently discriminative to reliably separate genuine cyst structures from other objects inside the retina.

To improve the spatial accuracy and reduce the dominant false positive error rate, future iterations of this pipeline can focus on replacing the basic features set by integrating locational context (where the pixels are in the retina) or textural features(patterns/structures). Furthermore, K-Means can be replaced with a more flexible unsupervised method or a better retina (ROI) segmentation method can be implemented.

References

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[3] G. N. Girish, B. Thakur, S. R. Chowdhury, A. R. Kothari, and J. Rajan, "Segmentation of Intra-Retinal Cysts From Optical Coherence Tomography Images Using a Fully Convolutional Neural Network Model," *IEEE Journal of Biomedical and Health Informatics*, vol. 23, no. 1, pp. 296–304, Jan. 2019, doi: 10.1109/JBHI.2018.2810379

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